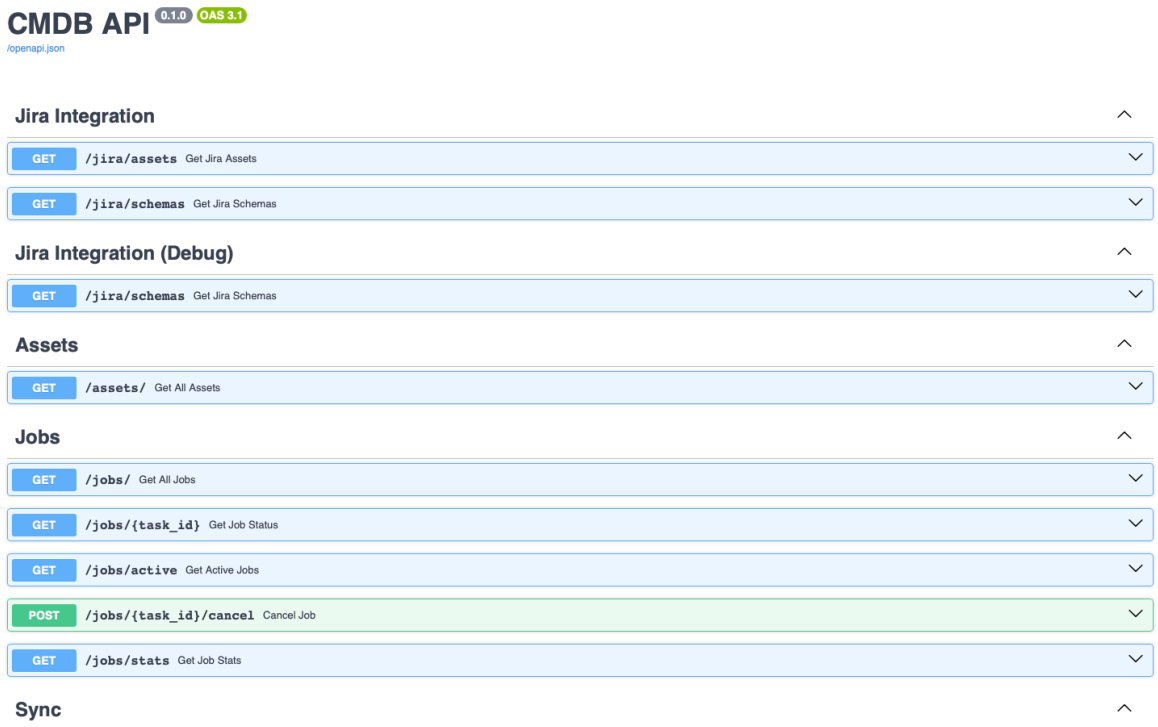
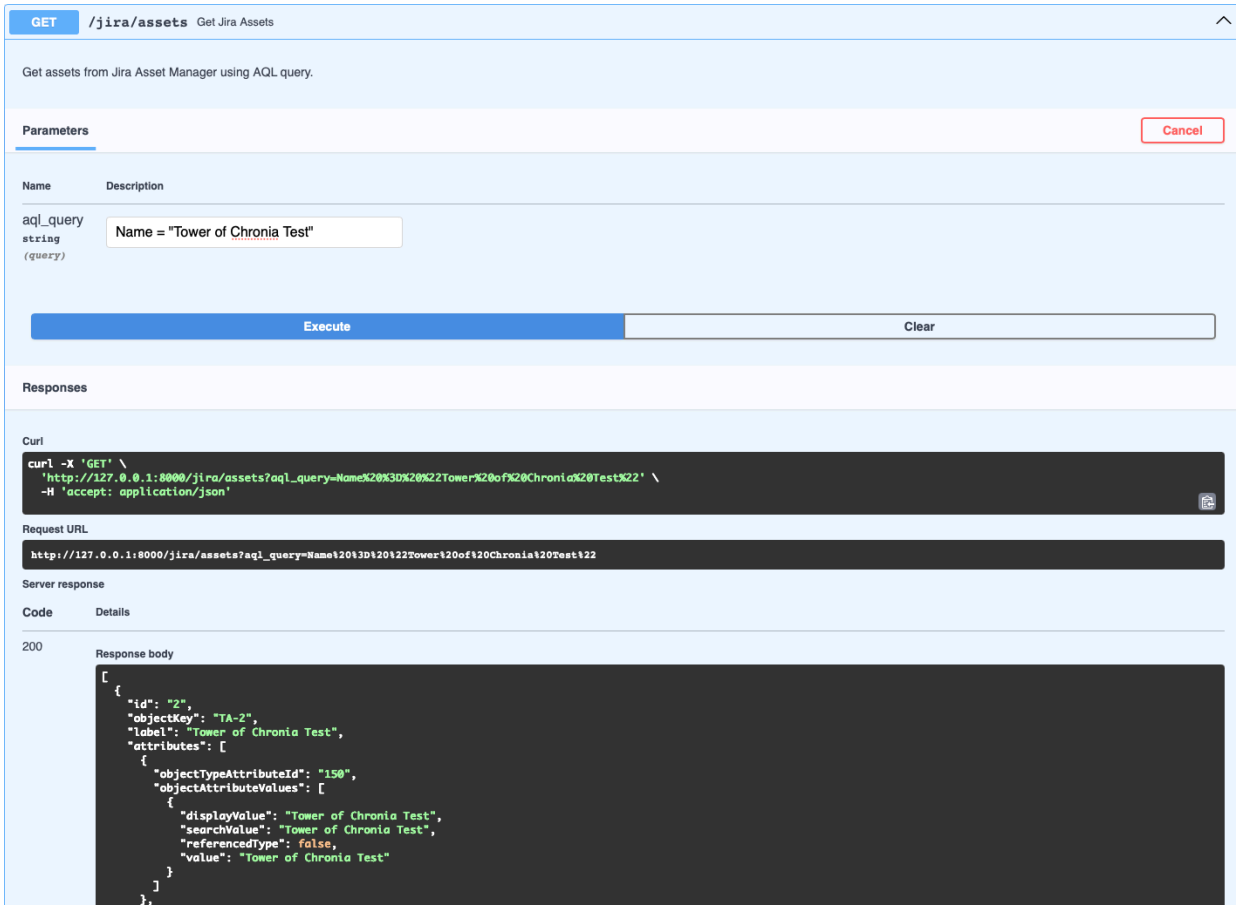
Application functionality.

Our project's core use case, Scheduled Automated Data Synchronization, is implemented and working. The system uses a scheduler to automatically trigger jobs for worker that discovers infrastructure assets with Ansible and then synchronizes that data with our Jira CMDB. Also, the conditions for UC-02 are implemented.

To run our project you need to have Linux machine/environment. To test it, run main.py file and test it via localhost:8000/docs.



Here we can see the GET api call on the server asset in Jira named ‘Tower of Chronia Test:



For the last release, we plan to have a .bat file for running the whole project.

Testing

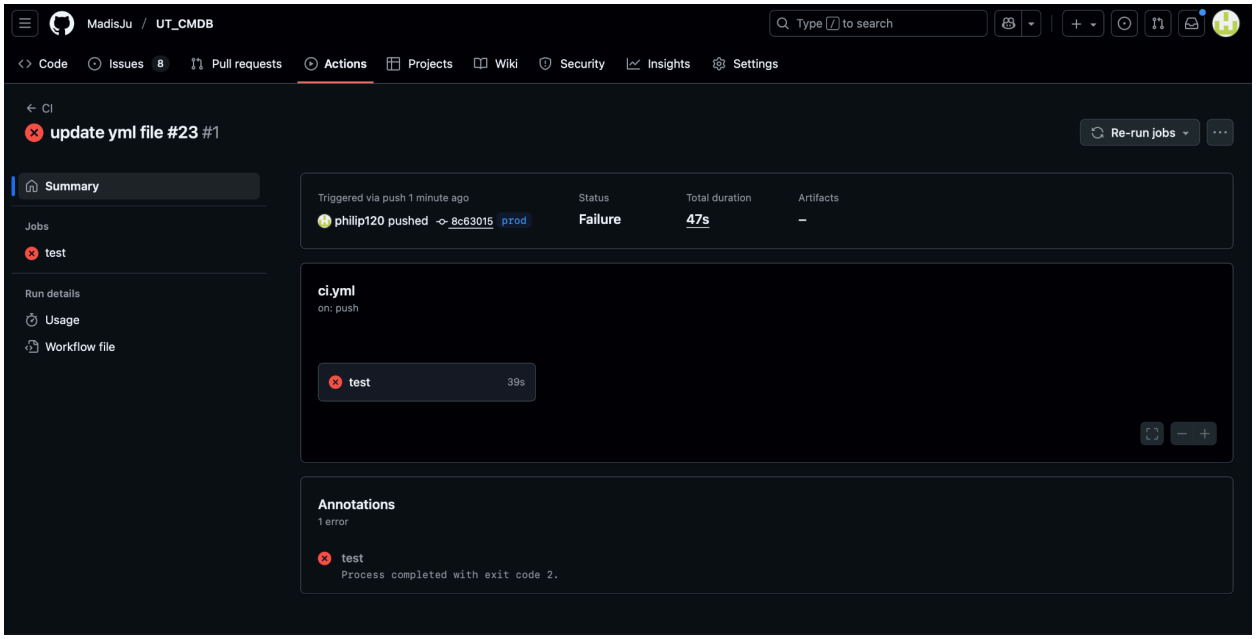
Initial project has test regarding the scheduler work, if it is assigning at least 1 task to worker, then the test is considered as passed. We use module called pytest:

```
(venv) philippaskov@Philips-MacBook-Pro UT_CMDB % pytest tests/test_scheduler.py
========================================= test session starts =========================================
platform darwin -- Python 3.11.9, pytest-8.4.2, pluggy-1.6.0
rootdir: /Users/philippaskov/Kool/semester5/softwareproject/UT_CMDB
plugins: anyio-4.11.0
collected 2 items

tests/test_scheduler.py ..                                                                  [100%]

========================================== warnings summary ===========================================
src/core/configs/config.py:7
  /Users/philippaskov/Kool/semester5/softwareproject/UT_CMDB/src/core/configs/config.py:7: PydanticDepr
ecatedSince20: Support for class-based `config` is deprecated, use ConfigDict instead. Deprecated in Py
dantic V2.0 to be removed in V3.0. See Pydantic V2 Migration Guide at https://errors.pydantic.dev/2.12/
migration/
    class Settings(BaseSettings):

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
================================== 2 passed, 1 warning in 0.07s ===================================
```

Continuous Integration (CI).

Continuous Integration is used throughout the process. Everyone has their own branch on Github and we are continuously committing and merging branches. Since we do not deploy our project somewhere, we do not use continuous deployment.

Continuous Integration is used in Github actions, the code is built and also tested there.

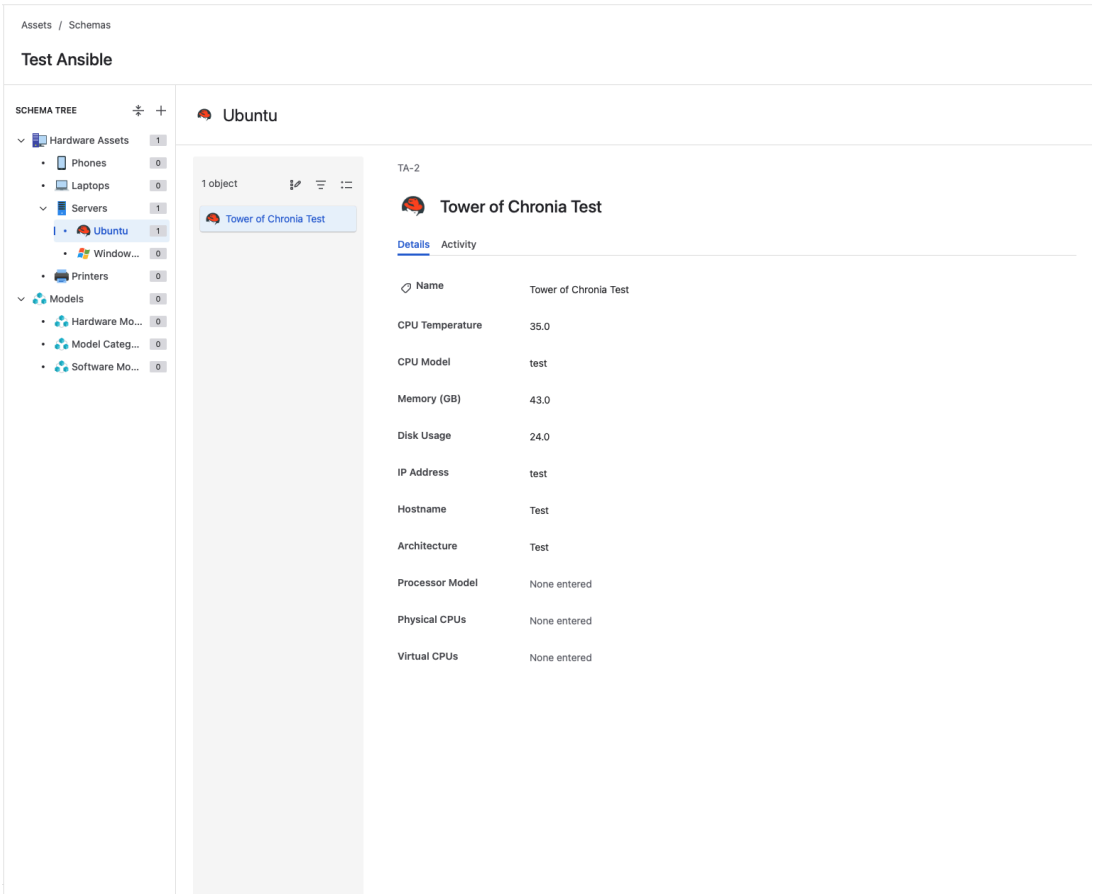


Detailed requirements.

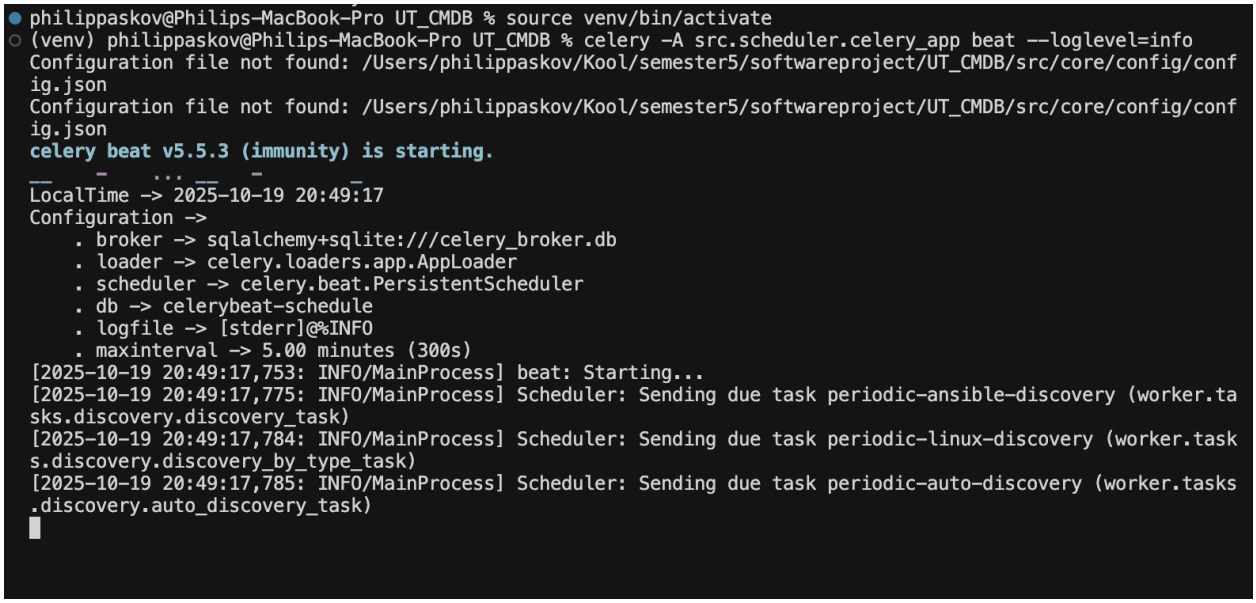
Available in Github Wiki.

UI wireframes/mockups.

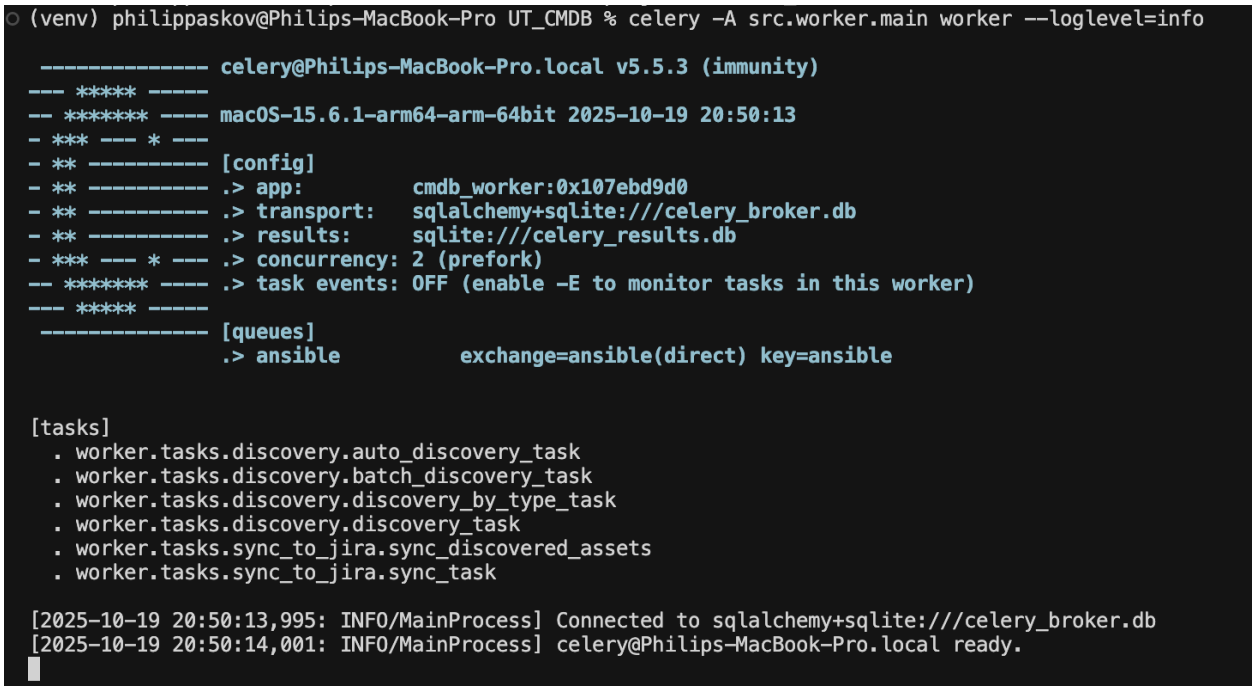
Jira Service Management (our UI) object “Tower or Chronia test” in schema “Test Ansible”:



Scheduler work:



Worker work:



Scope

- **Iteration 3:**
  More functionality
  Add support for Windows and Solaris systems
  Implement asset change tracking and audit logs

Expand test coverage and validation logic

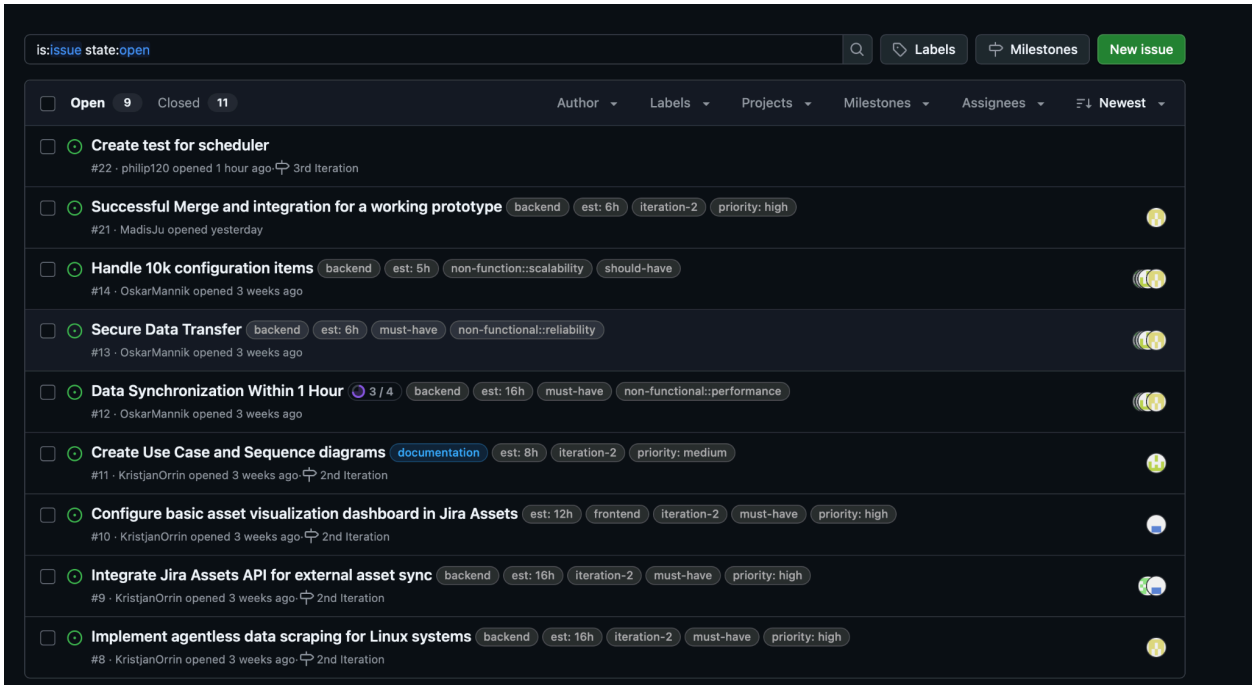
- **Iteration 4:**
  Finalization and Delivery
  Optimize performance and security
  Finalize documentation and architectural diagrams
  Conduct user testing and gather feedback
  Prepare deployment package and handover materials

Release notes.

Available in Github. https://github.com/MadisJu/UT_CMDB/releases/tag/release-2.0

Collaboration infrastructure is in use.

VCS is being used regularly and by all team members. Requirements in wiki or other documents are updated regularly and the issue tracking system is also updated and used regularly.



Individual contribution report

Madis - worker with core/ folder and overall logic of the project, connected different parts together

Philip - built Scheduler and its connections to other parts

Kristjan - built API part and worked with Jira

Oskar - built Worker and its connections to other parts